Whole numbers only

Optimization · Level 2 · Lesson 3



In this lesson you will

- Explain why some choices come only whole, or yes or no.
- Compare a quick (greedy) rule with the best load.
- Show why rounding can break a limit.

Key ideas

- Whole numbers: no half buses
- Yes or no: take it or leave it
- Greedy: fast, not always best
- Rounding can break a limit

Activity: Beat the quick rule

1. Cards (4 min): Six cards with weights and points.
2. Greedy (4 min): Pack the most points per kilo first.
3. Search (8 min): List loads that fit; find the best.
4. Compare (4 min): Did greedy miss? Who set the points?

Weights are in hundreds of kilos so the sums stay small: the truck holds 10.

Let’s discuss

1. Why did the solar panels make the quick rule miss?
2. Where else does rounding cause trouble?
3. Who should set the points when you help another school?

Exit ticket

- One choice that must be whole, and what rounding would get wrong.